

Ejercicio 2:

Hacer en lápiz y papel y subir un PDF con la resolución.

Dado $h[n] = \delta[n] - \delta[n-4]$, encontrar $y[n] = x[n] * h[n]$ para cada una de las siguientes $x[n]$:

a) $x[n] = \cos(\omega_0 n T_s)$. Expresar la respuesta como un único coseno de la forma $A \cos(\omega_0 n T_s + \phi)$.

b) $x[n] = (1/2)^n u[n]$.

c) $x[n] = u[n+1] - u[n-2]$.

$$h[n] = \delta[n] - \delta[n-4] \rightarrow y[n] = x[n] * h[n] / x[n]:$$

$$\text{Ec. (1)} \quad y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] \rightarrow \text{Convolve p/ un sist. LTI}$$

$$\text{Ec. (2)} \quad x[n] * \delta[n-n_0] = x[n-n_0] \rightarrow \text{Convolver con cualquier señal d/ u } \delta \text{ desplaza. NO deforma la señal, sólo la corre en el } t \text{ la } = \text{ cent. de neces.}$$

En general:

(aplicable a todos los incisos)

$$y[n] = x[n] * (\delta[n] - \delta[n-4]) = x[n] * \delta[n] - x[n] * \delta[n-4]$$

prop. distrib.

aplica Ec. (2)

$$y[n] = x[n] - x[n-4]$$

$$\text{(a)} \quad x[n] = \cos(\omega_0 n T_s)$$

$$y[n] = x[n] - x[n-4] = \cos(\omega_0 n T_s) - \cos(\omega_0 (n-4) T_s)$$

Prop. trig.: Resta de cosenos

$$\cos(\alpha) - \cos(\beta) = -2 \sin\left(\frac{\alpha+\beta}{2}\right) \sin\left(\frac{\alpha-\beta}{2}\right)$$

CAUX's:

$$\frac{\alpha+\beta}{2} = \frac{\omega_0 n T_s + \omega_0 (n-4) T_s}{2} = \frac{\omega_0 T_s (2n-4)}{2} = \omega_0 T_s n - 2\omega_0 T_s$$

$$\frac{\alpha-\beta}{2} = \frac{\omega_0 n T_s - \omega_0 (n-4) T_s}{2} = \frac{4\omega_0 T_s}{2} = 2\omega_0 T_s$$

$$\Rightarrow y[n] = \cos(\omega_0 n T_s) - \cos(\omega_0 (n-4) T_s) = -2 \sin(\omega_0 T_s n - 2\omega_0 T_s) \sin(2\omega_0 T_s)$$

cte., no dep. de n

$$\begin{aligned} \text{Prop. trig.} \\ -\sin(\theta) &= \cos\left(\theta + \frac{\pi}{2}\right) \\ \sin(\theta) &= \cos\left(\theta - \frac{\pi}{2}\right) \end{aligned}$$

$$y[n] = 2 \sin(2\omega_0 T_s) \cos(\omega_0 T_s n - 2\omega_0 T_s + \frac{\pi}{2})$$

$$A = 2 \sin(2\omega_0 T_s)$$

$$\Phi = -2\omega_0 T_s + \frac{\pi}{2}$$

$$(b) x[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$y[n] = x[n] - x[n-4] = \left(\frac{1}{2}\right)^n u[n] - \left(\frac{1}{2}\right)^{n-4} u[n-4]$$

func. escalón de pules.

$$u[n-c] = \begin{cases} 0 & \text{si } n < c \\ 1 & \text{si } n \geq c \end{cases}$$

$$(c) x[n] = u[n+1] - u[n-2]$$

$$y[n] = x[n] - x[n-4] = u[n+1] - u[n-2] - (u[n+1-4] - u[n-2-4])$$

$$= u[n+1] - u[n-2] - u[n-3] + u[n-6]$$